Prototype provided in collaboration with Alex S.



Figure 5 – Prism Protect logo

How big of a problem does colourblindness among university students pose?

- Statistically speaking, in a class of 30 people there will be at least one colourblind person.
- According to our data;

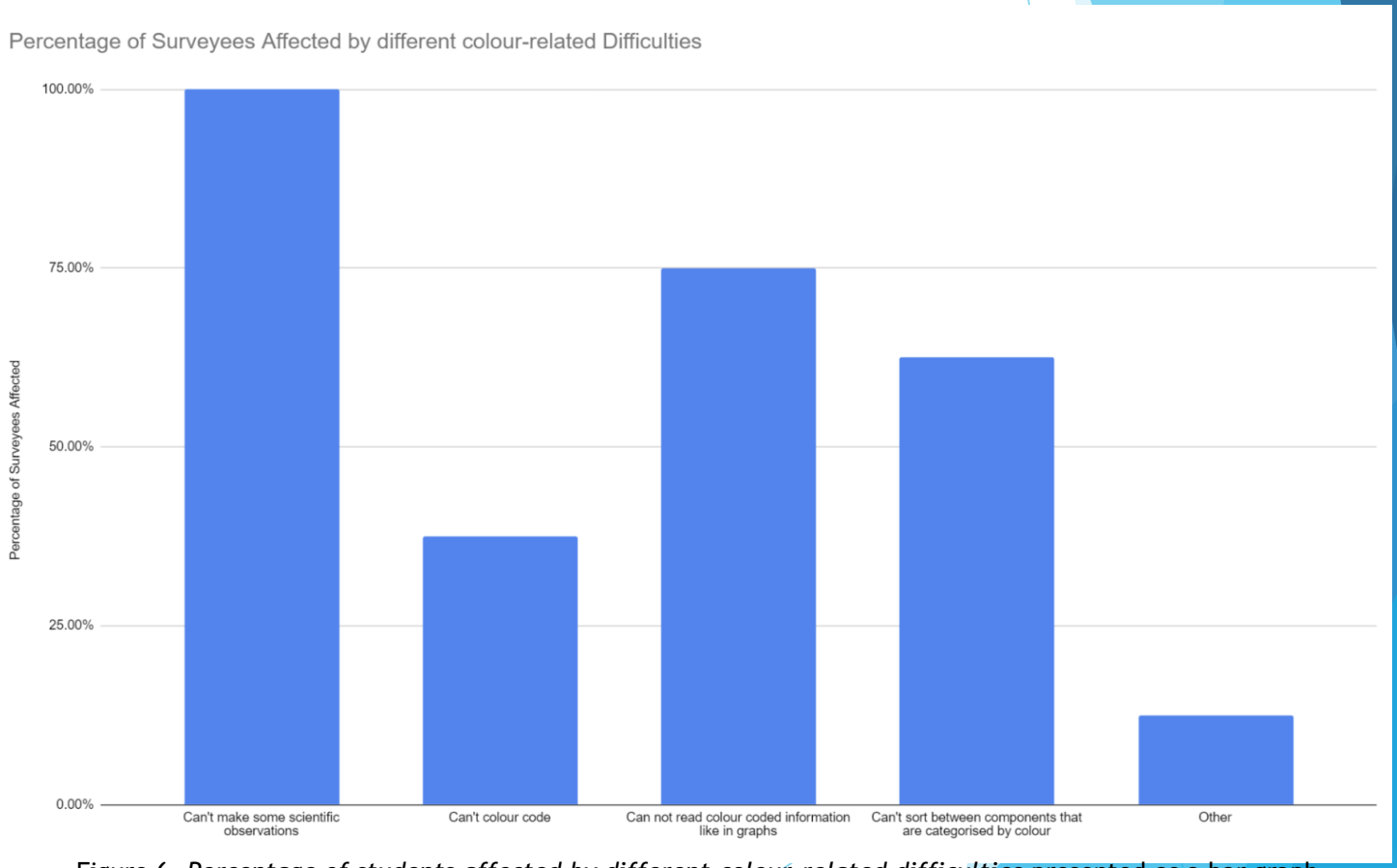


Figure 6. *Percentage of students affected by different colour-related difficulties* presented as a bar graph from Scoping Report (2023).

Our findings

62.5% of students were unable to continue their studies without effective assistance. (Scoping Report, 2023).

62.5% of respondents claimed that colour-corrective technology was something that they would be willing to spend money on. (Scoping Report, 2023).

100% of respondents sought assistance from a friend or lab partner with observations. (Scoping Report, 2023).

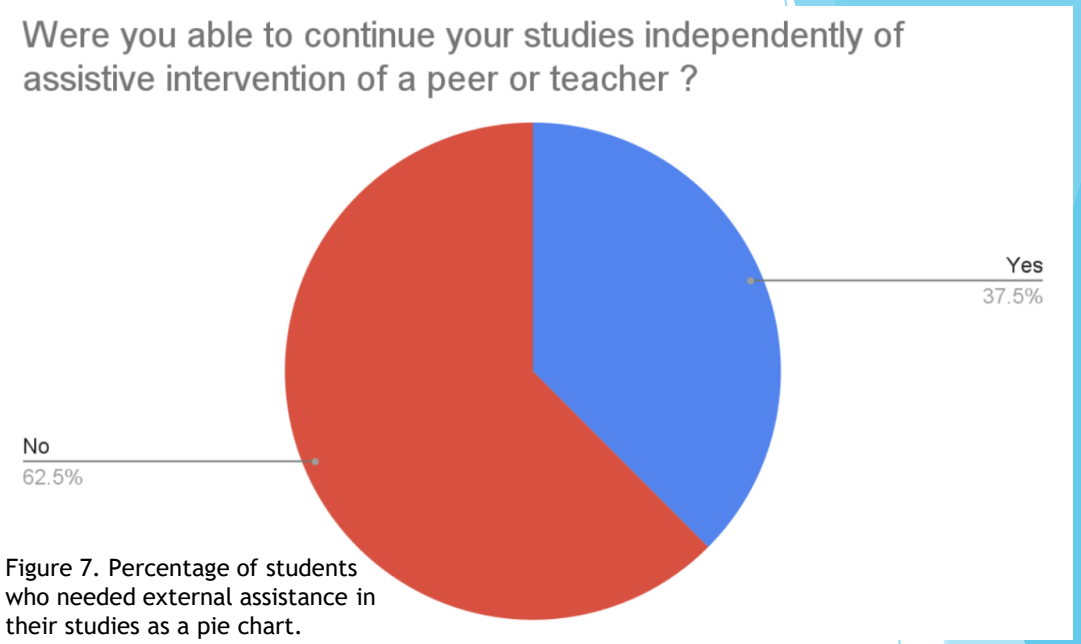


Figure 7. Percentage of students who needed external assistance in their studies as a pie chart.

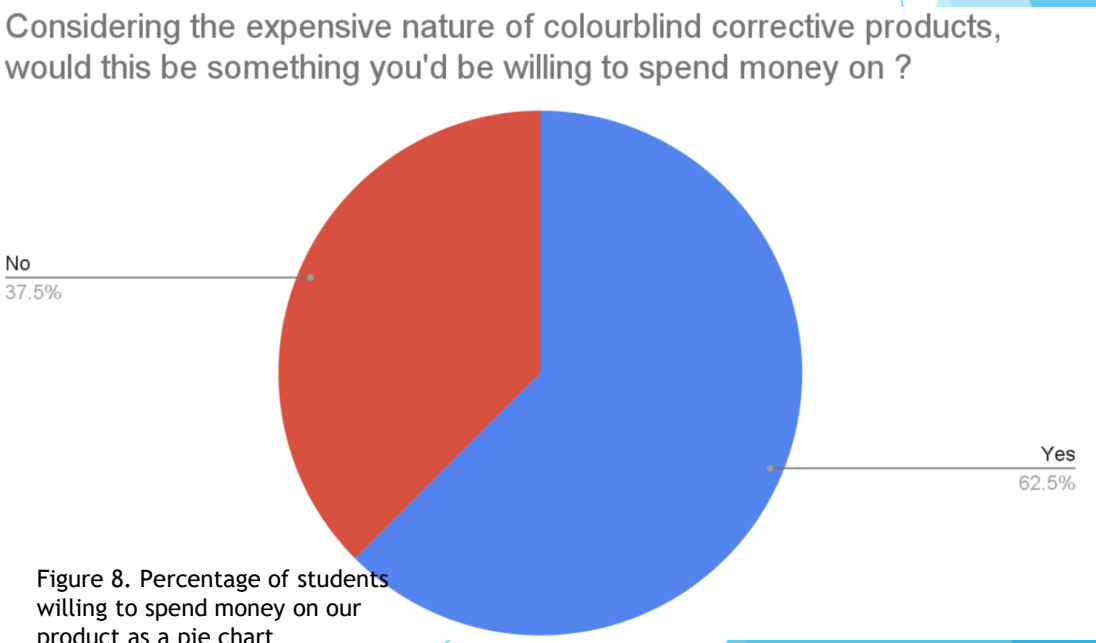


Figure 8. Percentage of students willing to spend money on our product as a pie chart

Our findings cont.

- No response mentioned receiving any corrective measures.
- Educators and facilitators regularly do not have accessibility measures for colour-blind students:
  - *"I don't think we've had any students with this condition in the past 3 years that we've run this course... we'd be very supportive of alternative arrangements to optimise the learning experience for them, should a case arise."* – J. J. ITBE Subject Coordinator. (Scoping Report, 2023)

- This called for an innovative solution to students struggling with colour-blindness in their classes.

Scope

PROBLEM

" Students with colour-blindness are unable to participate in lab activities that require colour observation, which makes it difficult to pass their classes independently "

SOLUTION

Affordable, safety compliant, modular, colour corrective glasses accessible by students, teachers, and facilitators

Competition Analysis

1. **EnChroma**

- Colour-corrective filters proved ineffective on users ability to differentiate colours:
  - *"EnChroma filters had no significant improvement." (Almutairi et al., 2017)*
  - *"The results show that the [Enchroma Inc] glasses... neither improve results in the diagnosis tests NOR allow the observers with color blindness to have a more normal color vision."* (Gómez-Robledo et al., 2018)
- Average cost of $300-$600+, could be out of budget range for students
  - However, 62.5% of survey responders given nature of cost of colour-blind glasses, said they would purchase this product (Scoping Report, 2023)
- Frame design made of lightweight, sturdy material, 'TR-90', however, it is not suitable as PPE

2. **VINO Optics** – biggest competitor

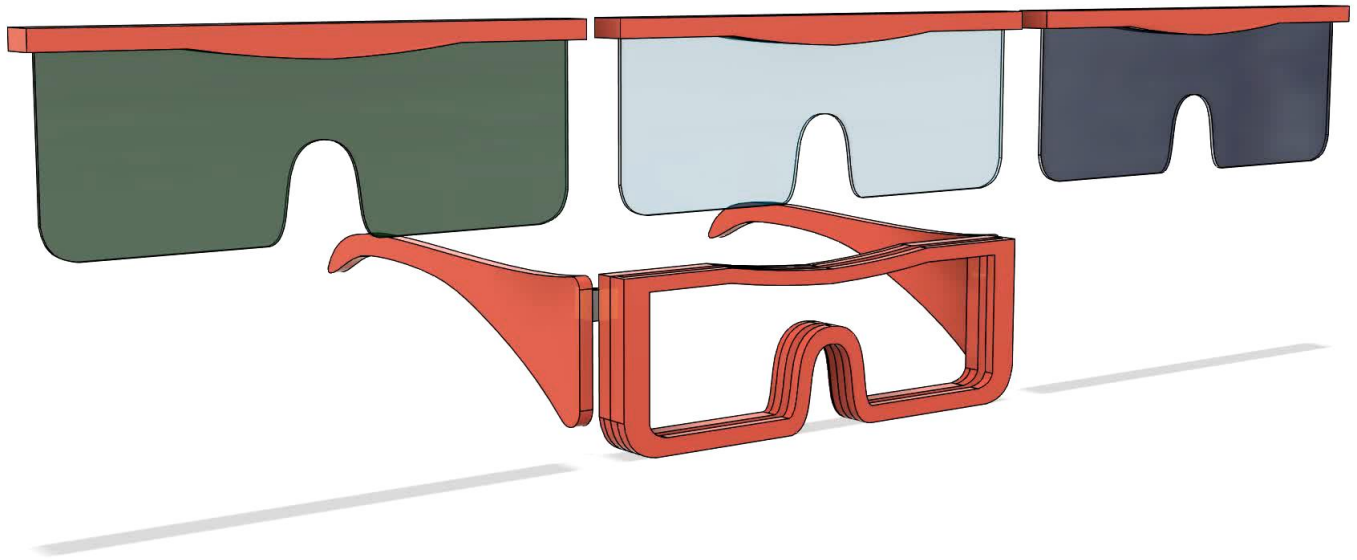
- Protective colour-corrective glasses meets AS/NZS 1337 standards
- Modular: lenses can be replaced and clicked on and off frames
- Lenses proven to be effective in creating colour discrimination
  - *"Discrimination IMPROVED along the RG axis both for deuteranopic (p <.001) and deuteranomalous (p =.05) (red/green) subjects wearing VINO OPTICS (O2Amp) glasses."* (Mastey et al., 2016)
- Not as costly as EnChroma, however, still costly
  - Subscription plans costing $250-$1200 a year with warranty benefits
  - Single glasses ranging from $300-$400, and prescription grade lenses costing $100
- Caters to red-green colour-blindness **only**

We're not worried!

- We're better in:
  - Modularity
  - Affordability
  - Safety Compliance

Innovation 1 - Modularity

Figure 9 – 3D animation showing slide interchangeability



- Interchangeable colour corrective filter lenses offer versatility to students and facilitators
- Proper care and storage of unused lenses should be exercised to minimise the risk of damages

Innovation 2 - Affordability

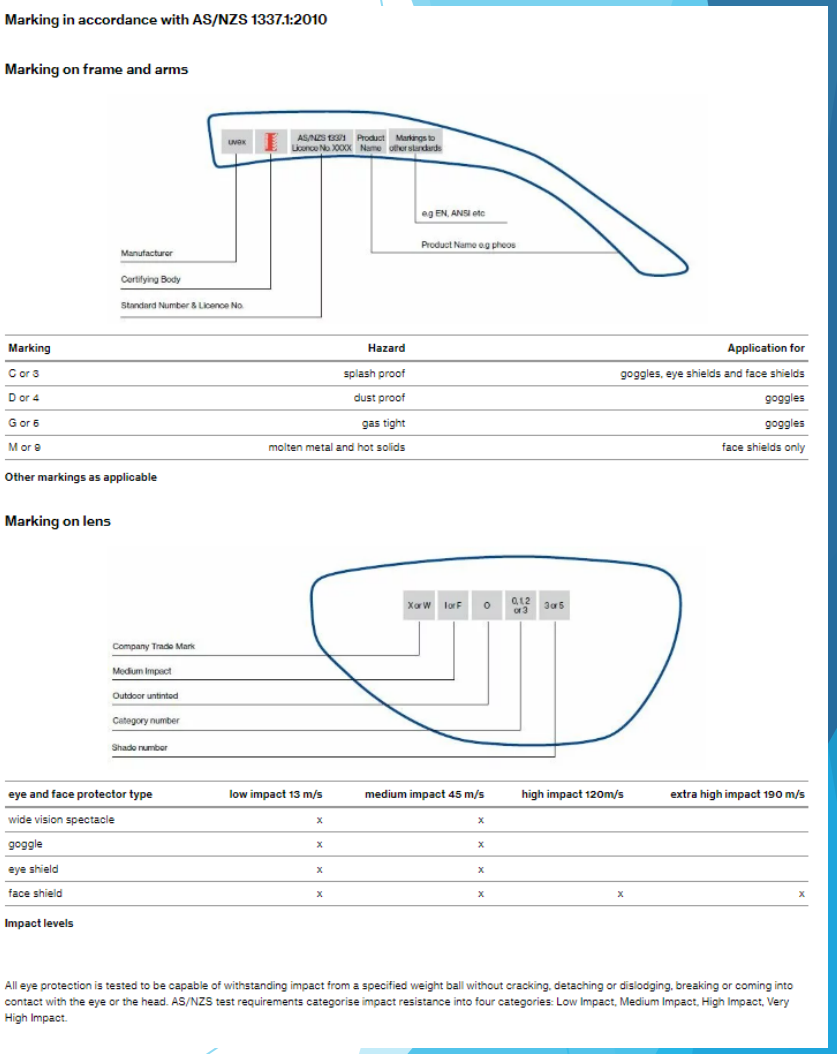
- Achieved through efficient manufacturing techniques (3D printing and CNC cutting)
- Since the construction is partitioned and then assembled, the durability and shock resistance of the glasses may not perform as well as other glasses
- Slides cost at most $100 AUD to source and manufacture
- Frame materials cost $3 - $10 a pair + $10 hinges
- Manufacturing Equipment
- Retail costs $150 and $220



Figure 10 – 3D animation showing disassembly and reassembly of glasses

Innovation 3 - Safety Compliance

- Compliance with Uvex (2023) specified Australian and New Zealand Personal Protective Equipment Standards.
  - Materials
  - Manufacturing
  - Design



Marking in accordance with AS/NZS 1337.1:2010

Marking on frame and arms

Marking	Hazard	Application for
C or 3	splash proof	goggles, eye shields and face shields
D or 4	dust proof	goggles
G or 6	gas tight	goggles
M or 9	molten metal and hot solids	face shields only

Other markings as applicable

Marking on lens

eye and face protector type	low impact 13 m/s	medium impact 45 m/s	high impact 120m/s	extra high impact 190 m/s
wide vision spectacle	x	x		
goggle	x	x		
eye shield	x	x		
face shield	x	x	x	x

Impact levels

All eye protection is tested to be capable of withstanding impact from a specified weight ball without cracking, detaching or dislodging, breaking or coming into contact with the eye or the head. AS/NZS test requirements categorise impact resistance into four categories: Low Impact, Medium Impact, High Impact, Very High Impact.

Figure 11 – Basic AN/NZS regulation codings Uvex (2023)

Material

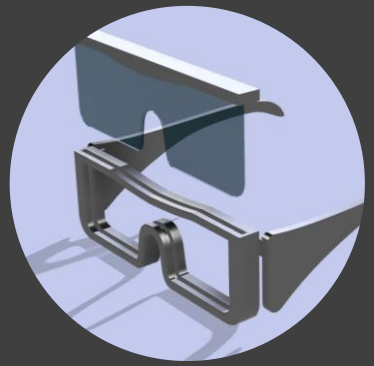
- Materials used in the model are utilised in the construction of existing safety glasses, for their strength, durability, impact resistance and compliance with Australian and New Zealand Safety Standards (Uvex, 2023)

Aluminum

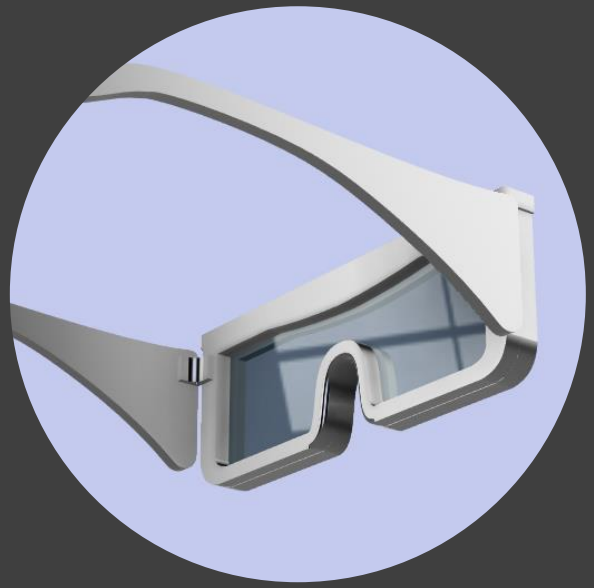
- Aluminium is more easily recycled than other materials because it is strong, durable, flexible, impermeable, lightweight, and corrosion-resistant.since Aluminum is of the most recyclable metals. (Moore,2022)

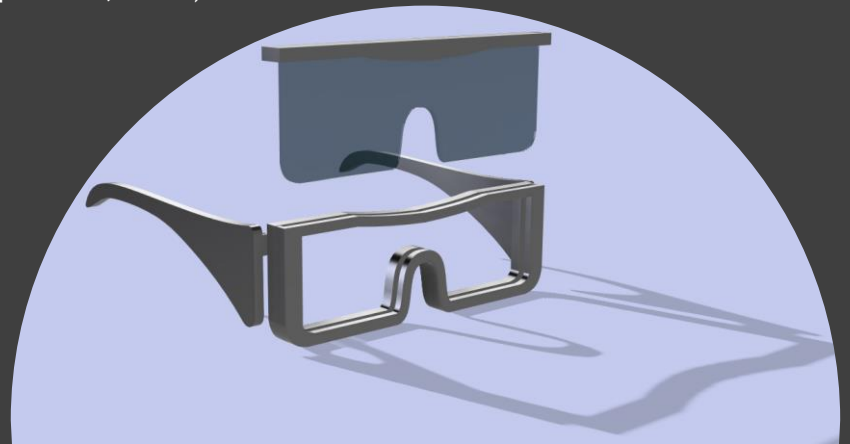
Plastic

- PLA, which stands for Polylactic Acid
- PLA plastic is biodegradable and so is environmentally friendly in the case that it is disposed of - such as shavings of PLA (Australasian Bioplastics, 2019)











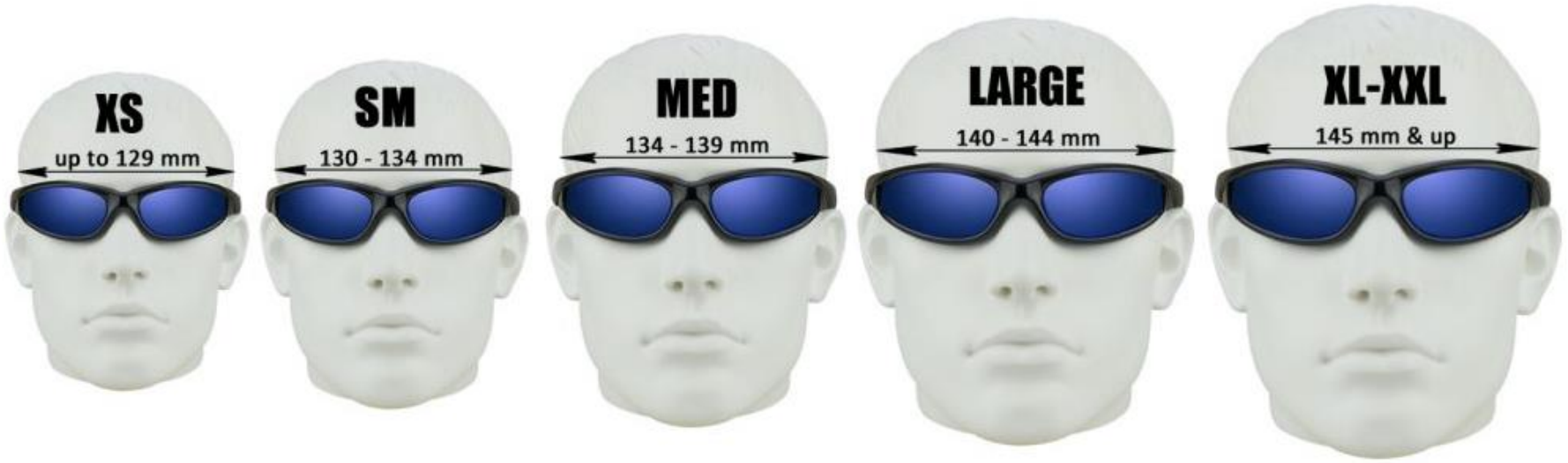


Figure 12 – Generic image showing glasses sizes. Glasses sizes determined using RoShamBo Eyewear (2022)

Size and Repairability

- The use of flat and simple parts are to be assembled using simple processes makes the frame highly modular, easy to assemble and disassemble.

We've all heard of a colourful future, so why not see it clearly with Prism-Protect?

REFERENCES

- Australasian Bioplastics. (2019). *Bioplastics explained*. ABA Australasian Bioplastics Association. https://bioplastics.org.au/bioplastics/bioplastics-explained
- Australian Government . (2018, March 4). *Aluminium*. Australian Mineral Facts. https://www.ga.gov.au/education/classroom-resources/minerals-energy/australian-mineral-facts/aluminium
- Colour Blind Awareness. (2022, April 27). *About Colour Blindness - Colour Blind Awareness*. Colour Blind Awareness. https://www.colourblindawareness.org/colour-blindness
- EnChroma. (2018, April 24). *EnChroma - Optique*. Optique. https://optiqueiowa.com/enchroma
- Gómez-Robledo, L., Valero, E. M., Huertas, R., Martínez-Domingo, M. A., & Hernández-Andrés, J. (2018). Do EnChroma glasses improve color vision for colorblind subjects? *Optics Express*, *26*(22), 28693. https://doi.org/10.1364/oe.26.028693
- Hawley, D. (2019, April 10). *Comfort For The First Time Eyeglasses Wearer*. Maple Grove Eye Doctors | Pearle Vision. https://maplegroveeye.vision/comfort-for-the-eyeglasses-neophyte
- Mastey, R., Patterson, E. J., Summerfelt, P., Luther, J., Neitz, J., Neitz, M., & Carroll, J. (2016). Effect of "color-correcting glasses" on chromatic discrimination in subjects with congenital color vision deficiency. *Investigative Ophthalmology & Visual Science*, *57*(12), 192–192. https://iovs.arvojournals.org/article.aspx?articleid=2558387
- Moore, K. (2022, September 13). *Everything You Need To Know About Recycled Aluminium | Capral Aluminium*. Capral Aluminium. https://www.capral.com.au/blog/news/everything-you-need-to-know-about-recycled-aluminium
- Optimem. (2018). *Australian Standards for Safety Glasses*. Safety Glasses Online. https://safetyglassesonline.com.au/australian-standards-for-safety-glasses
- PriceWaterhouseCoopers. (2021). *Consumers becoming more eco-conscious and willing to travel, attend mass events, PWC survey reveals*. PWC. https://www.pwc.com/gx/en/news-room/press-releases/2021/global-consumer-insights-pulse-survey-2021.html
- RoShamBo Eyewear. (2022). *Sunglasses Size Chart | Glasses Size Chart by Age*. Ro•Sham•Bo Eyewear. https://www.roshambo.com/pages/sunglasses-size-chart
- Scoping Report. (2023). *Scoping Report*. Google Docs. https://docs.google.com/document/d/1zem4B7lDMMm0T2CuPWfBTmNwp5fgpgSd-MJhiWVMtZE/
- Uvex. (2023). *Safety Eyewear | Australian Standard AS/NZS 1337.1 | uvex Safety Australia*. Uvex-Safety.com.au. https://www.uvex-safety.com.au/en/knowledge/safety-standards/safety-glasses